Walnut Technologies™

PRODUCT PORTFOLIO · IoT PAYMENT DEVICES

Smart Payment End-to End Ecosystem

Customised Design | Multi Language | TMS + MQTT

Real-time audio payment confirmation for QR-based commerce — engineered, designed and manufactured in-house. Configurable for global standards and deployments.

Your partner for IoT Solutions



Walnut Technologies™

PRODUCT PORTFOLIO · IoT PAYMENT DEVICES

Our Clientele



paytm



HDFC BANK



mswipe

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Our Clientele



SBI



BharatPe



airtel Payments Bank



MobiKwik™



Easebuzz

and many others...

THE OPPORTUNITY

Audio confirmation is now table-stakes

QR payments surged in popularity more rapidly than the hardware supporting them. Merchants require a quick, hands-free method to verify each transaction without needing to look at their phones. The soundbox addressed this issue effectively. We went a step further by creating a family of devices that are designed, manufactured, and software-controlled in-house, catering to markets beyond India.

100% of payments confirmed by voice — no screen, no staff dependency

< 2 sec transaction-to-announcement delay on the live network

4 form factors from pocket device to keypad terminal — one platform



Truly made in India — for the future digital payments network of the world

FLAGSHIP DEVICE

Soundbox — Core Specifications



ABS shell
HB fire rating
1.0 m drop-tested
BIS certified

Processor
Dual-core ARM Cortex-R, high-performance module

Memory
8 MB onboard

Connectivity
4G (LTE Cat-1), optional 4G + 2G fallback

Audio
3W-5W · 93–95 dB

Battery
2000–2600 mAh (optional), Type-C charging

Security
SSL / TLS 1.2 secure connectivity, OTA updates

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PRODUCT FAMILY

One platform, four form factors

Shared core electronics and a common TMS backbone — differentiated by display, input and portability to fit every merchant needs.



Soundbox

Audio-first QR device

RGB LED · 4 keys · 3W speaker



Pocket Soundbox

Portable, on-the-go merchant

1000 mAh · ultra-compact



DQR 4.3"

Dynamic QR + colour display

4.3" LCD · merchant display



All -in-One Soundbox

Tap, DIP/Chip insert and Dynamic QR

4'3 . LTE CAT 1 – 4G, 2G, BT, Wifi

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PRODUCT FAMILY

Featuring - AI based Soundbox

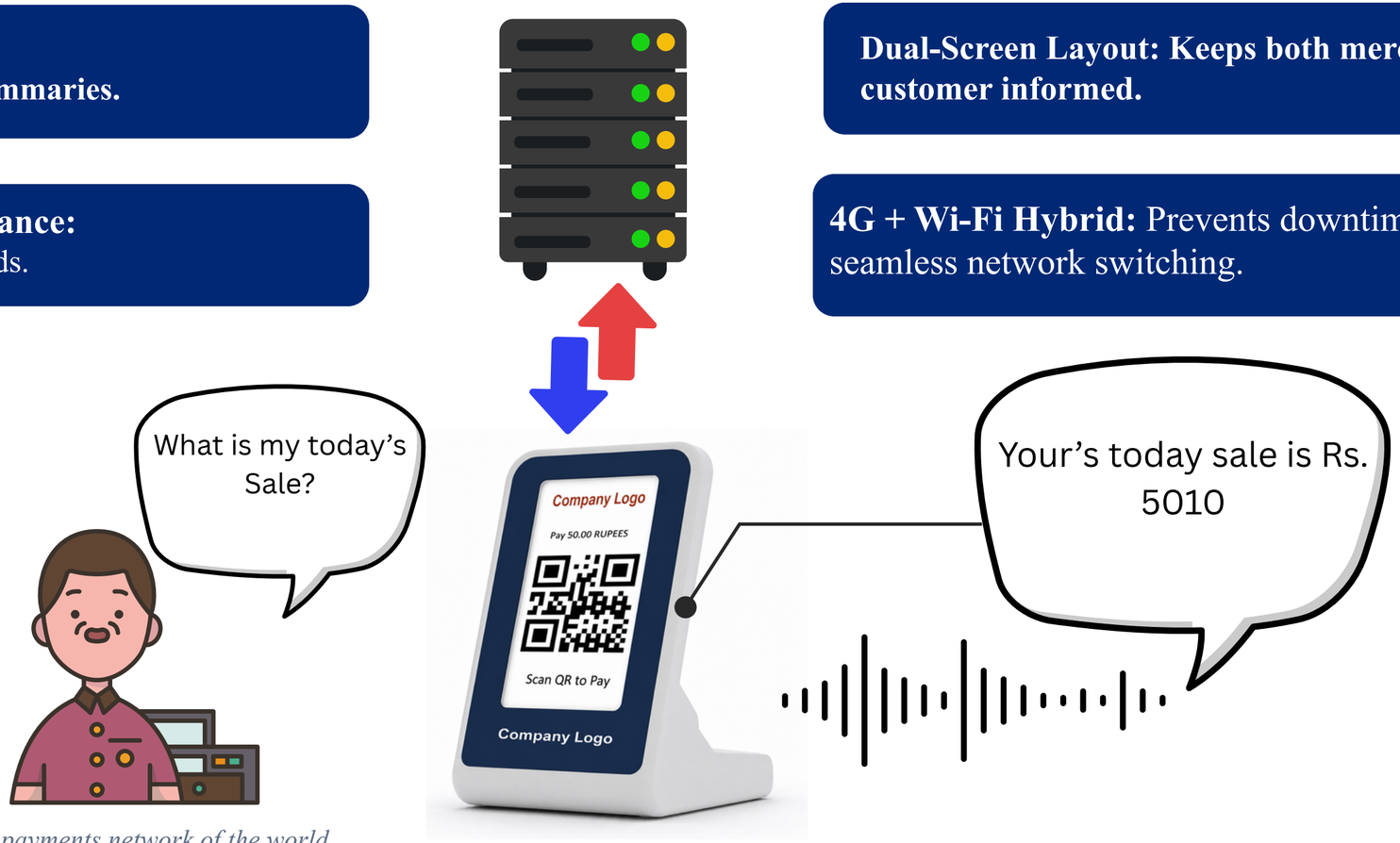
The **AI Soundbox** is a next-generation merchant terminal that transforms simple payment alerts into intelligent business management. By combining reliable audio confirmations with a voice-activated AI assistant, it acts as a digital manager for retail counters.

Voice AI Assistant:
Gives instant spoken business summaries.

Omnichannel Payment Acceptance:
Speeds up queues with QR and cards.

Dual-Screen Layout: Keeps both merchant and customer informed.

4G + Wi-Fi Hybrid: Prevents downtime with seamless network switching.



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THE WALNUT EDGE

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Designed, built and coded under one roof

We control the entire stack—from the enclosure CAD and PCB to firmware, cloud, and the management console. This capability allows us to swiftly adapt to new markets, compliance requirements, and brands.

In-house R&D Design

Industrial design, enclosure CAD and PCB layout developed by our own engineering team — not sourced as a reference board.

Own Manufacturing Factory

Assembly, flashing and quality control on a controlled line. ABS shell, 1.0 m drop-tested, BIS-certified output.

Core Technology

Dual-core ARM Cortex-R module, 4G/LTE Cat-1 connectivity, SSL/TLS security and OTA-ready firmware architecture.

Own Firmware and TMS

Firmware, MQTT broker, AWS IoT pipeline and the TMS console — every layer engineered and maintained internally.

OUR VARIANTS

Compare the family at a glance

Specifications	Soundbox	Pocket	DQR 4.3"	All-in-One
Processor	ARM Cortex-R	ARM Cortex-R	ARM Cortex-R	ARM Cortex-R
Display	—	—	4.3" colour LCD	4.3" colour LCD
Merchant display	—	—	2" × 0.5"	2" × 0.5"
Input	4 keys	4 keys	Numeric Keypad	Numeric keypad
Speaker output	93-95 dB	≈ 75 dB	93-95 dB	93-95 dB
Audio PA	3-5W	2W	3-5W	3-5W
Battery	2000–2600 mAh	1000 mAh	2000–2600 mAh	2000–2600 mAh
Connectivity	4G + opt. 2G	4G (LTE Cat-1)	4G + opt. 2G	4G + opt. 2G
Best for	Counter, audio-first	Mobile merchant	Display + audio	Full terminal

SOFTWARE SOLUTION FOR LARGE SCALE DEPLOYMENT

TERMINAL MANAGEMENT SYSTEM — MANAGE YOUR FLEET ANYWHERE, ANYTIME

A control centre for your entire terminal estate. TMS streamlines the full device lifecycle — from deployment to OTA updates and troubleshooting — with real-time visibility into every unit.

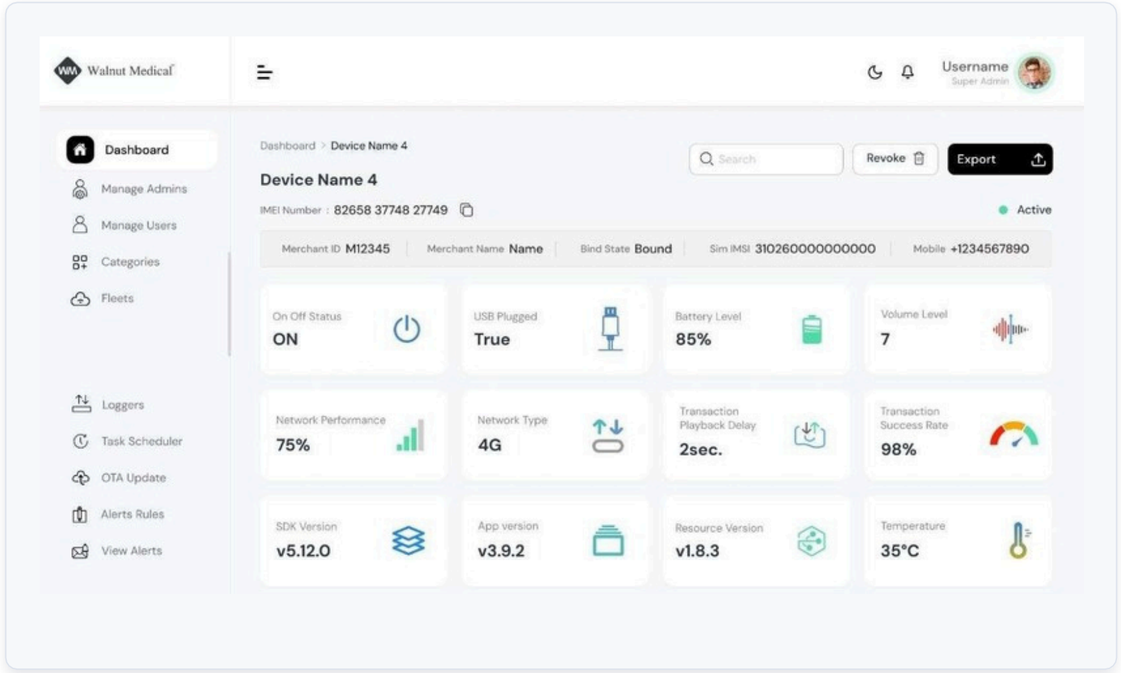
Device & network Storage, CPU, memory, battery, IP, IMEI, operator & live location.

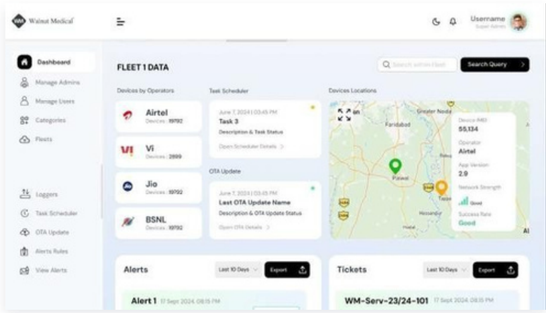
App details App, SDK, sound-file & language versions; install history.

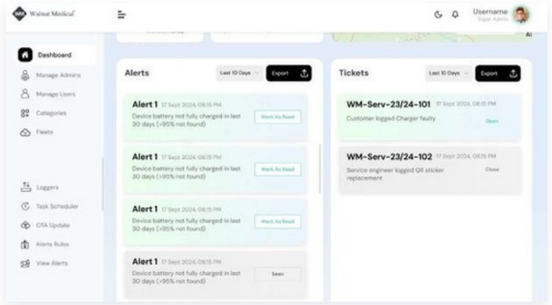
Scheduler Schedule OTA jobs, advertisements and remote commands.

Onboarding Link to merchant accounts, release OTP, un-bind, watch txn matrix.

Defender Security metrics, anomaly detection, audits, alerts & mitigation.







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FULL TELEMETRY

Walnut Technologies™

40+ device parameters, tracked in one go

Every device reports a rich live telemetry stream to TMS — so support, security and finance teams always see ground truth.

Health & hardware

- Online / offline status
- Battery level & temperature
- CPU, task & heap memory
- External storage size / used / free
- USB plugged state
- Audio volume level

Network & SIM

- Network strength & type
- SIM operator & details
- IMEI / IMSI numbers
- Mobile number
- MCC / MNC codes
- Time zone

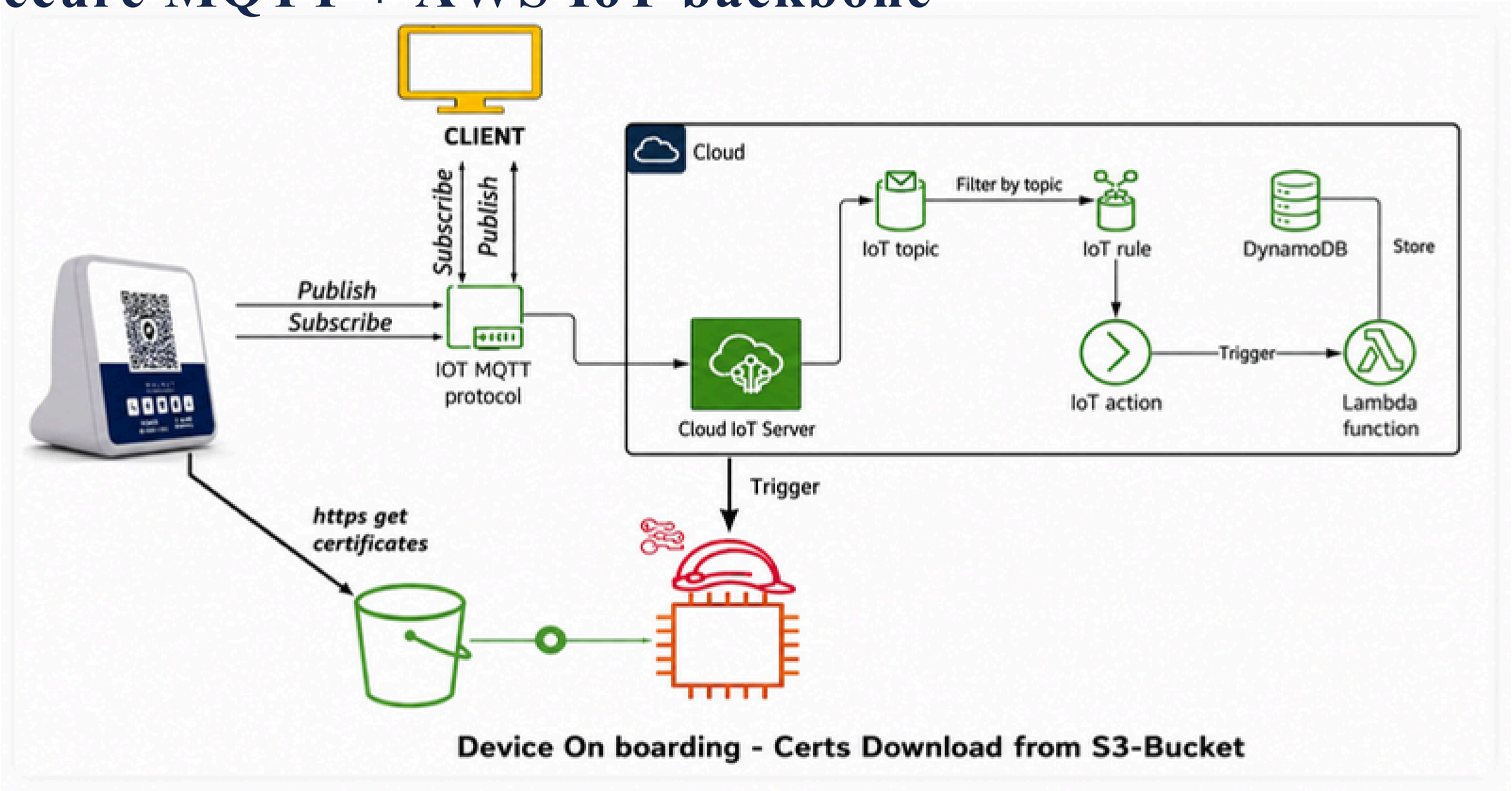
Device Details

- Device model & first-online
- SDK / app / file versions
- Bind state & merchant
- Bound QR code
- Switch-on time

Transactions

- Transaction success rate
- Audio play delay
- Per-device txn matrix
- Live activity monitoring

Secure MQTT + AWS IoT backbone

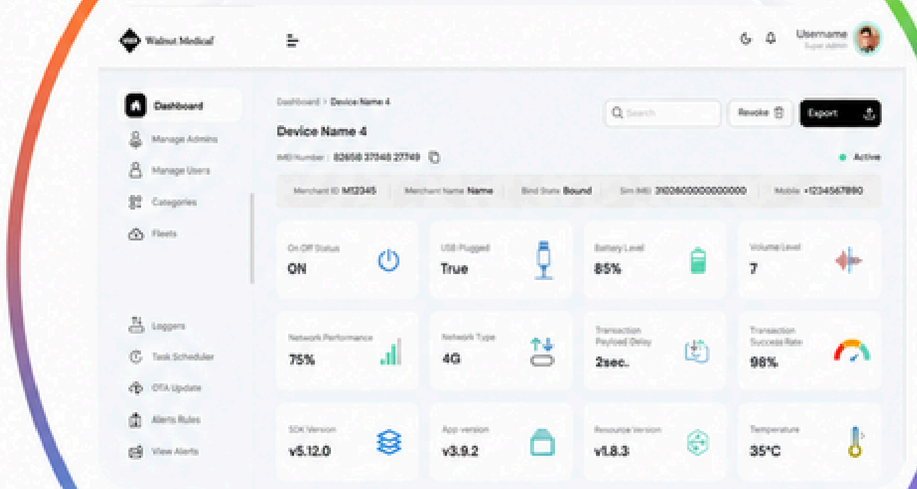


Dedicated connection

MQTT traffic isolated on its own path for low latency and minimal interference.

TLS 1.2 encryption

End-to-end encryption, MITM protection, and verified data integrity in transit.



Centralised broker

Single broker — simpler config, access control and policy enforcement at this scale.

Lightweight by design

No distributed cluster overhead; intentionally manageable for current reliability needs.

DEPLOYMENT

What we handle for an integration

01

Payment trigger

Wire your bank / PSP events into the device via webhook, WebSocket, SSE, message queue or pub-sub.

02

AWS configuration

Device configuration and load balancing tuned on AWS for efficient, reliable operation at scale.

03

Merchant onboarding

A flexible onboarding flow adapted to your KYC, account-linking and OTP requirements.

04

API integration

Clean APIs for software integration — communication and data exchange between your systems and ours.

05

Database schema

A well-defined schema to structure, store and manage device and transaction data effectively.

06

Ongoing support

OTA firmware, advertisement scheduling, remote commands and direct engineering support.

Walnut Technologies™

BUILD ON A PLATFORM YOU OWN

Let's bring audio payments to your market.

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